

Section 4: The Cisco IOS Operating System

Layer: Multi | Reference: Odom Vol1 Ch4: Using the Command-Line Interface | *Original study notes — CCNA 200-301*

The Cisco IOS command-line interface (CLI) is the primary way to configure and verify Cisco routers and switches. Understanding its modes, memory locations, and navigation shortcuts is foundational - nearly every other CCNA topic assumes fluency here.

Key Concepts

- User EXEC mode (Router>): limited to basic show/troubleshooting commands, no configuration access
- Privileged EXEC mode (Router#): full show/debug access and the gateway to configuration mode; entered via 'enable'
- Global Configuration mode (Router(config)#): entered via 'configure terminal'; changes affect the whole device
- Interface Configuration mode (Router(config-if)#): entered via 'interface <type><slot/port>'; changes apply only to that interface
- Line Configuration mode (Router(config-line)#): entered via 'line vty/console <range>'; controls terminal-line behavior like login and transport
- 'exit' moves up one mode level; 'end' or Ctrl-Z jumps directly back to Privileged EXEC from any depth
- Running-config lives in RAM and is the currently active configuration - lost on reboot if not saved
- Startup-config lives in NVRAM and is loaded automatically at boot
- IOS image itself is stored in Flash memory; ROM holds the bootstrap program and a minimal ROMMON/rescue IOS
- 'copy running-config startup-config' (or 'write memory') saves the active config so it survives a reboot
- Tab completes a partially typed command; '?' shows context-sensitive help; Up-arrow recalls command history
- 'show running-config | include <text>' or '| begin <text>' filters long output to relevant lines
- 'enable password' stores a weak, reversible password; 'enable secret' uses an MD5 hash and always takes precedence if both are configured
- 'service password-encryption' applies weak Type 7 encryption to other plaintext passwords in the config (not a strong protection, but better than plaintext)

Command Reference

Command	Purpose
enable	Move from User EXEC to Privileged EXEC mode
configure terminal	Enter Global Configuration mode from Privileged EXEC
exit / end	exit: up one level. end: jump straight to Privileged EXEC
show running-config	Display the active configuration currently in RAM
show startup-config	Display the configuration saved in NVRAM

Command	Purpose
copy running-config startup-config	Save the active config to NVRAM (same as 'write memory')
hostname <name>	Set the device's hostname (Global Config mode)
no shutdown	Administratively enable an interface
do show <command>	Run any Privileged EXEC show command from within configuration mode
service password-encryption	Apply weak Type 7 encryption to plaintext passwords in the config
enable secret <password>	Set an MD5-hashed privileged-mode password (preferred over enable password)

Configuration Walkthrough

Navigate from boot to interface config and back

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface GigabitEthernet0/0
R1(config-if)#no shutdown
R1(config-if)#end
R1#
```

Save configuration and back it up to TFTP

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
R1#copy run tftp
Address or name of remote host []? 10.10.10.10
```

Secure the enable password properly

```
R1(config)#enable secret StrongPass123
R1(config)#service password-encryption
```

Worked Scenario

Q: A device was configured and saved correctly yesterday, but after a power outage overnight it booted up with none of yesterday's changes. What happened?

Someone (or a script) never ran 'copy running-config startup-config' after making the changes. Running-config lives only in RAM and is lost on any reboot or power loss; only what's saved to startup-config in NVRAM survives. The fix going forward: always save explicitly after verified changes, and consider automated config backups to a TFTP/SCP server.

Common Mistakes / Gotchas

- Confusing 'enable password' (weak, reversible) with 'enable secret' (MD5 hash) - if both exist, enable secret wins, but leaving enable password configured is still a bad practice
- Forgetting 'no shutdown' is required on most interfaces by default (interfaces come up administratively down)
- Assuming 'service password-encryption' provides strong security - Type 7 is trivially reversible and only deters casual viewing
- Typing 'end' when meaning to go up only one level, accidentally exiting configuration entirely

Exam Tips

- ★ *Know exactly where each config lives: RAM (running) vs NVRAM (startup) vs Flash (IOS image) vs ROM (bootstrap)*
- ★ *enable secret always overrides enable password when both are present*
- ★ *Practice rapid mode-switching commands - simlets often require navigating several mode levels in one task*